Yun Seok Kim

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Research Interests

Algorithm–hardware co-design for efficient generative AI, ML accelerator architecture, and memory-centric computing systems.

Education

Yonsei University — Seoul, Republic of Korea
M.S. in Electrical and Electronic Engineering (GPA: 3.96/4.3) — *Mar. 2025 – Present (Expected Feb. 2027)*

Sungkyunkwan University — Suwon, Republic of Korea
B.S. in Electronic and Electrical Engineering (GPA: 3.97/4.5) — *Mar. 2023 – Feb. 2025*

Inha University — Incheon, Republic of Korea
Undergraduate studies in Electronic Engineering (GPA: 3.92/4.5) — *Mar. 2018 – Feb. 2023*

- Transferred to Sungkyunkwan University.

Research Experience

Graduate Researcher, Yonsei University — *Mar. 2025 – Present*
Embedded Systems and Computer Architecture Lab
Advisor: Prof. Won Woo Ro

- Co-developed **Autobahn**, a token reduction algorithm and dedicated accelerator for autoregressive video generation. Exploited reference-image and temporal redundancy with position-aligned comparisons, and designed token compaction and position restoration hardware, achieving up to **3.41× speedup** over a systolic-array baseline without token reduction. [*In preparation*]
- Conducted motivation experiments on **KV cache retrieval for autoregressive image generation**, analyzing attention patterns and retrieval overhead to characterize the tradeoffs between inference latency and generation quality. [*In preparation*]
- Investigated **memory-centric LLM inference**, beginning with FlashAttention computation and data movement on reconfigurable PIM architectures. Extended the study to multiple GDDR6-PIM devices, comparing tensor and pipeline parallelism to examine how workload mapping and device topology affect computation and communication costs. [ICCE-Asia 2025]

Publications *Denotes equal contribution.

Manuscripts

- **Yun Seok Kim***, Sungbin Kim*, Mingu Jung, Enhyeok Jang, and Won Woo Ro. "Autobahn: Efficient Autoregressive Video Generation Based on Semantic-Temporal Reduction.", 2026. [preprint]

Conference Papers

- **Yun Seok Kim** and Won Woo Ro. "TAPAS: A Case Study on Topology-Aware Multi-PIM in Model Parallelism." *International Conference on Consumer Electronics Asia (ICCE-Asia)*, Busan, Korea, 2025. [link]
- Hoje Lee, **Yun Seok Kim**, and Won Woo Ro. "Real-Time Access Pattern-Based Page Migration Policy in CXL Memory Systems." *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Jeju, Korea, 2025.

Technical Skills

- **Frameworks:** CUDA, Nsight Systems, Nsight Compute, PyTorch
- **Programming Languages:** C/C++, Python, Verilog

Scholarships and Awards

Fellowships and Scholarships

- BK (Brain Korea) 21 Research Fellowship, *Yonsei University* — *2025–26*
- Graduate Student Research Assistant (GSRA), *Yonsei University* — *2025*

Awards

- Outstanding Presentation Award, *Automotive System IC Fusion Human Resource Research Center (ASRC)* *2026*

Teaching Experience

Teaching Assistant, Yonsei University

- EEE 4473: Embedded System Lab. (Undergraduate) *Spring 2025*
  - Contributed lectures on programmable logic and processing systems using Zynq 7020 kit.
  - Organized practice questions and had Q&A sessions during lectures.

Contributed Talks

- **Advanced LLM Systems: Agentic AI**
  - Large Language Model (LLMs) Seminar, *Yonsei University* *Feb. 2026*
- **Understanding Multi-GPU Programming**
  - Advanced GPU Seminar, *Yonsei University* *Aug. 2025*
- **Advanced Branch Prediction and Dynamic Scheduling**
  - Advanced Computer Architecture Seminar, *Yonsei University* *Feb. 2025*

Service and Affiliations

- Graduate Student Member, *Institute of Electrical and Electronics Engineers (IEEE)* *2025–26*
- Student Volunteer, *IEEE/ACM International Symposium on Microarchitecture (MICRO)* *2025*
- Trainee, *Automotive System IC Fusion Human Resource Research Center (ASRC)* *2025–26*
- Battalion Commander Interpreter (KATUSA), *1-38 FAB, 2nd Infantry Division, 8th US Army* *2020–22*

Language Proficiency

Korean (Native), English (TOEFL iBT: 5.0 / 6.0)